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# Network Analysis

It Is techuique used for planning and scheduling. large projects In field of Construction, maintanance fabrication, purchasing & app design etc..

In this large project are broken down to individual activities and are arranged In a logical sequence.

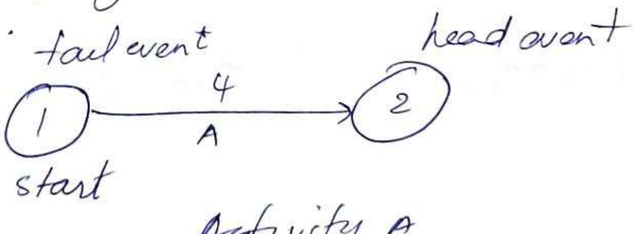
Network analysis helps In designing, planning, Coordinating and ~~prei~~ decision making. Inorder to accomplish the project economically. In the minimum available time, with the limited available resources.

The most popular modern techniques are. CPM - critical path method. PERT → Program evaluation and review technique.

Terms related to PERT & CPM

Event : Is a speific Instant of time that Indicates the beginning or end of an activity

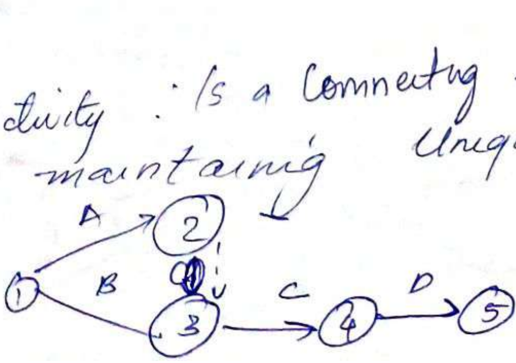
Activity : Is any type / resource Conservaing path of the project which has a definite start & finish.




Activity A

Activities are classified as Critical & non critical and. dummy.

dummy Activity : Is a Connecting link for Control purpose or for maintaining Uniqueness of Activity.



Network or Arrow diagram : A pictorial representation of a project plan showing the Interrelationship b/w the various activities in a sequences

slack or float : a margin of extra time over & above its duration which a non Critical activity can consume without delaying the date of Completion of project.

$EFT = EST + duration$

(EFT → Earliest finish time; EST → Earliest start time)

$LST = LFT - duration$

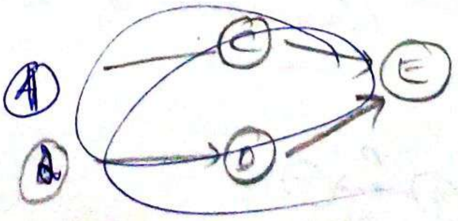
(LST → Latest start time; LFT → Latest finish time)

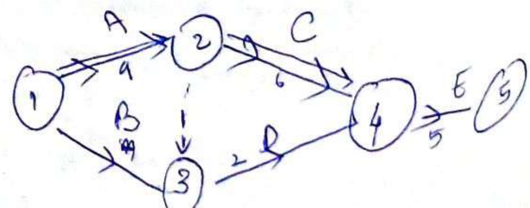
Float / slack = Time available for Completion − Time neccessary to complete.

Critical path : A path along the network in which earliest finish time and latest finish time are equal ~~it true~~ it is that sequence of activities which decides the total project duration and Consume max time. It has zero float.

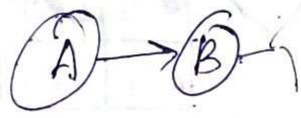
① Draw the network diagram for the project in which pre operations and post operations are given below

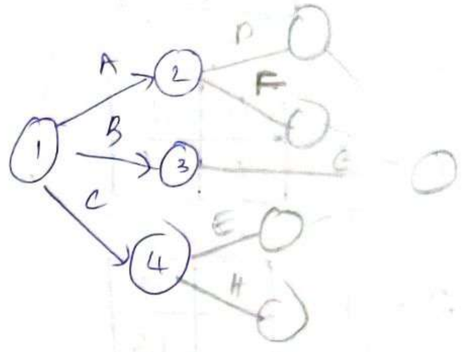
| OPT | Pre OPT | Post OPT |
|---|---|---|
| A | — | C |
| B | — | D |
| C | A | E |
| D | B | E |
| E | C D | none |



| OPT | PreOPT | Post |
|---|---|---|
| A | – | DF |
| B | – | G |
| C | – | EH |
| D | A | G |
| E | C | G |
| F | A | – |
| G | ABCDE | – |
| H | C | – |



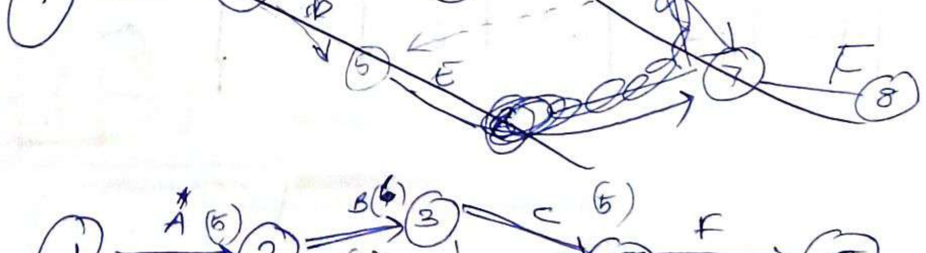


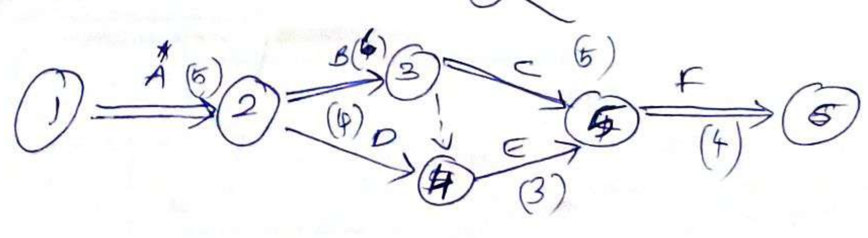
A small scale Industrial Unit Consist of 6 activities as given below.

| Activity | timing Days | Preopt. | EST | EFT | LST | LFT |
|---|---|---|---|---|---|---|
| A | 5 | None | 0 | 5 | 0 | 5 |
| B | 6 | A | 5 | 11 | 5 | 11 |
| C | 5 | B | 11 | 16 | 11 | 16 |
| D | 4 | A | 5 | 9 | 9 | 13 |
| E | 3 | D | 9 | 12 | 13 | 16 |
| F | 4 | C,E | 16 | 20 | 16 | 20 |

Draw the network diagram & Calculate EST, LST, EFT, LFT & float & mark the Critical path and find the total project duration.

To Prove Critical path float is zero





| | |
|---|---|
| EST | EFT |
| LST | LFT |

float = LFT − EFT

or EFT − LST.

1−2

| | |
|---|---|
| 0 | 5 |
| 0 | 5 |

0 0 — critical

float

Consider least (In EST & LFT)

2 − 3

| | |
|---|---|
| 5 | 11 |
| 5 | 11 |

0 0 — critical.

2 − 4

| | |
|---|---|
| 5 | 9 |
| 9 | 13 |

3 − 5

| | |
|---|---|
| 11 | 16 |
| 11 | 16 |

4 − 5

| | |
|---|---|
| 9 | 12 |
| 13 | 16 |

5 − 6

| | |
|---|---|
| 16 | 20 |
| 16 | 20 |

0 0

Junction.
So take ~~smallest box~~ higher value (EST EFT)

Duration of Critical path

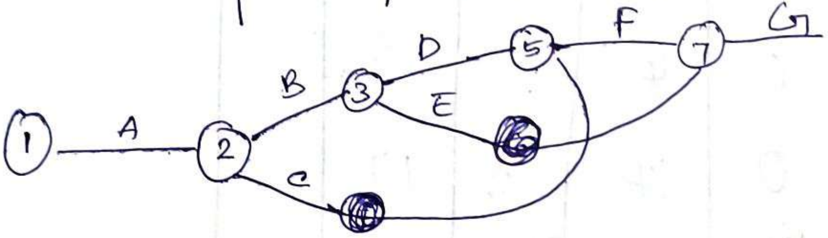
= 1 − 2 − 3 − 5 − 6

= 5 + 6 + 5 + 4 = 20

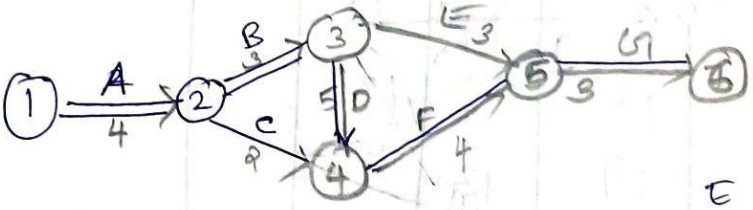
| Activity | Duration | EST | EFT | LST | LFT | Total float | Remark |
|---|---|---|---|---|---|---|---|
| 1-2 A | 5 | 0 | 5 | 0 | 5 | 0 | 0 - CP |
| 2-3 B | 6 | 5 | 11 | 5 | 11 | 0 | 0 - CP |
| 3-5 C | 5 | 11 | 16 | 11 | 16 | 0 | 0 - CP |
| 2-4 D | 4 | 5 | 9 | 9 | 13 | 4 | N CP |
| 4-5 E | 3 | 9 | 12 | 13 | 16 | 4 | N CP |
| 5-6 F | 4 | 16 | 20 | 16 | 20 | 0 | 0 - CP |

1. The activity details ~~and~~ given below along with their activity times, Construct the network diagram.

| Activity | Preop | Time | EST | EFT | LST | LFT | Float | Remark |
|---|---|---|---|---|---|---|---|---|
| A | – | 4 | 0 | 4 | 0 | 4 | 0 | CP |
| B | A | 3 | 4 | 7 | 4 | 7 | 0 | CP |
| C | A | 2 | 4 | 6 | 10 | 12 | 6 | NCP |
| D | B | 5 | 7 | 12 | 7 | 12 | 0 | CP |
| E | B | 3 | 7 | 10 | 13 | 16 | 6 | NCP |
| F | C, D | 4 | 12 | 16 | 12 | 16 | 0 | CP |
| G | E, F | 3 | 16 | 19 | 16 | 19 | 0 | CP |




CP = 19 days




1 – 2
A

| | |
|---|---|
| 0 | 4 |
| 0 | 4 |

0 0

2 – 3
B

| | |
|---|---|
| 4 | 7 |
| 4 | 7 |

0 0

2 – 4
C

| | |
|---|---|
| 4 | 6 |
| 10 | 12 |

6

3 – 4
D

| | |
|---|---|
| 7 | 12 |
| 7 | 12 |

0 0

E
3 – 5

| | |
|---|---|
| 7 | 10 |
| 13 | 16 |

6

F
4 – 5

| | |
|---|---|
| 12 | 16 |
| 12 | 16 |

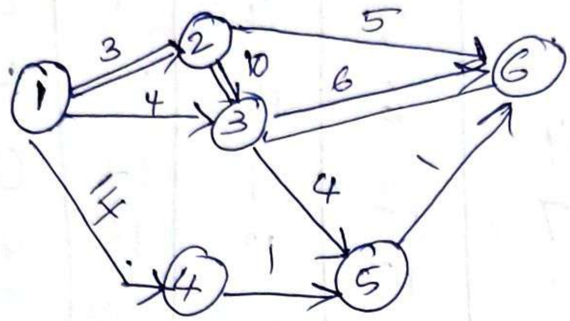
0 0

G
5 – 6

| | |
|---|---|
| 16 | 19 |
| 16 | 19 |

0 0

② Calculate EST & LFT, LST, EFT, TF, Project Duration for the network given in fig, Also calculate total project duration and show the Critical path on the network.




$CP = 1-2-3-6$
$3+10+6 = \underline{19}$

| Activity | time | EST | EFT | LST | LFT | F | Remark. |
|---|---|---|---|---|---|---|---|
| 1-2 | 3 | 0 | 3 | 0 | 3 | | |
| 1-3 | 4 | 0 | 4 | 1 | 3 | | |
| 1-4 | 14 | 0 | 14 | 3 | 17 | | |
| 2-6 | 5 | 3 | 8 | 14 | 19 | | |
| 2-3 | 10 | 3 | 13 | 3 | 13 | | |
| 3-6 | 6 | 13 | 19 | 13 | 19 | | |
| 3-5 | 4 | 13 | 17 | 14 | 18 | | |
| 4-5 | 1 | 14 | 15 | 17 | 18 | | |
| 5-6 | 1 | 17 | 19 | 18 | 19 | | |

②

1-2

| | |
|---|---|
| 0 | 3 |
| 0 | 3 |

2-3

| | |
|---|---|
| 3 | 13 |
| 3 | 13 |

0 0

5-6

| | |
|---|---|
| 17 | 18 |
| 18 | 19 |

1

1-3

| | |
|---|---|
| 0 | 4 |
| 1 | 3 |

3-5

| | |
|---|---|
| 13 | 17 |
| 14 | 18 |

3-6

| | |
|---|---|
| 13 | 19 |
| 13 | 19 |

0 0

1-4

| | |
|---|---|
| 0 | 14 |
| 3 | 17 |

4-5

| | |
|---|---|
| 14 | 15 |
| 17 | 18 |

2-6

| | |
|---|---|
| 3 | 8 |
| 14 | 19 |

Project duration

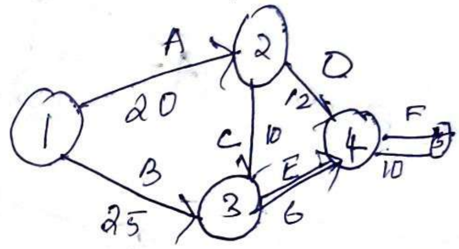
12/1/24

The Activities involved in a small project are given below along with relevant Information. Construct the network and find the Critical path

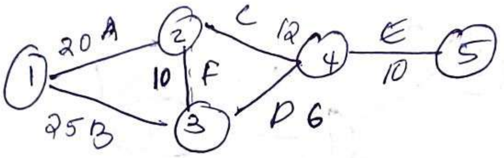
| Activity | Duration |
|---|---|
| 1 - 2 | 20 |
| 1 - 3 | 25 |
| 2 - 3 | 10 |
| 2 - 4 | 12 |
| 3 - 4 | 6 |
| 4 - 5 | 10 |

THIS IS CPM - Activity oriented

Correct Method of representation




| EST | EFT |
|---|---|
| LST | LFT |




1 - 2

| | |
|---|---|
| 0 | 20 |
| 0 | 20 |

1 - 3

| | |
|---|---|
| 0 | 25 |
| 5 | 30 |

5

2 - 3

| | |
|---|---|
| 20 | 30 |
| 20 | 30 |

2 - 4

| | |
|---|---|
| 20 | 32 |
| 24 | 36 |

4

3 - 4

| | |
|---|---|
| 30 | 36 |
| 30 | 36 |

4 - 5

| | |
|---|---|
| 36 | 46 |
| 36 | 46 |

LETUS plant to start a Computer Centre by procuring a room available In Ernakulam city the main activities are To procure the building and purchase Computers & Install it in the office. LETUS first the activities Involved In this project Without any logical ord

(B). Preparation of specification to purchase computers

(A) 2 To procure the available building in ernakulam city.

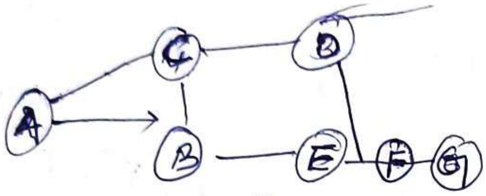
(D) 3) Install AC in room

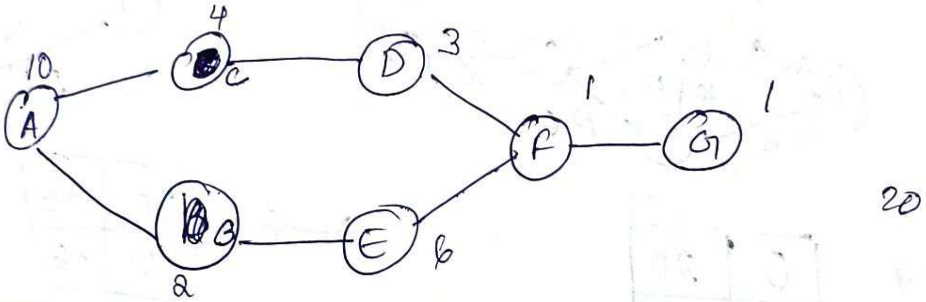
(C) 4 Set electrical connection and white wash (E) (5) purchase computers as per specification

(F) 6) Install computers

(G) 7) Test run the computers

* Draw the network diagram.




A - B

| 0 | 10 |
|---|---|
| | |

B - E

| | |
|---|---|
| | |

A - C

| 10 | 14 |
|---|---|
| | |

E - F

| | |
|---|---|
| | |

F - G

| | |
|---|---|
| | |

C - D

| | |
|---|---|
| | |

D - F

| | |
|---|---|
| | |

* PERT → Program Evaluation and Review technique,
→ ~~Progressive model~~. Probabilistic model
→ Event oriented.
→ 3 time estimate – $t_o$, $t_p$, $t_m$.
→ R&D or Defence projects (More Uncertainty projects.

$t_o$ → Optimistic time (shortest period of time to complete an activity)
$t_p$ → pessimistic time (Maximum period of time to complete)
$t_m$ → Most likely time (Best time, In b/w $t_o$ & $t_p$.)

* Critical path method (CPM) has only one time estimate so it does not consider uncertainty in time. PERT takes into account the uncertainty of activity times. it is a probabilistic model with uncertainty in activity duration. PERT make use of 3 estimates of time $t_o$, $t_p$, $t_m$.

$t_o$ : shortest possible time if every thing go perfectly without any complication, it is an estimate of minimum possible time to complete the activity under ideal conditions.

$t_p$ : it is the longest time, taking into consideration if everything goes wrong.

$t_m$ Best estimate of activity time, lies between $t_o$ & $t_p$ estimates.

The 3 time estimates are combined to develope Expected time ($t_e$).

$$t_e = \frac{t_o + 4t_m + t_p}{6}$$

Standard deviation $\sigma = \frac{t_p - t_o}{6}$

Variance $\sigma^2 = \left(\frac{t_p - t_o}{6}\right)^2$.

## Probablity of Completion of a project within a scheduled time

Probablity of Completion of a project is given by the probablity of occurrance of the end event of the network. The probablity distribution is assumed to be normal distribution. The standard normal variate for the desired time is given by-

$$Z = \frac{ts - te}{\sigma}$$

① A small engineering project Consist of 6 activities the 3 time estimates In nos of days for each activity are given below.

| Activity | to | tm | tp |
|---|---|---|---|
| 1 – 2 | 2 | 5 | 8 |
| 2 – 3 | 1 | 1 | 1 |
| 3 – 5 | 0 | 6 | 18 |
| 5 – 6 | 7 | 7 | 7 |
| 1 – 4 | 3 | 3 | 3 |
| 4 – 5. | 2 | 8 | 14 |

ⓐ Calculate the values of expected time, standard deviation. and Variance for each activity

ⓑ Draw the network diagram and mark te on each activity

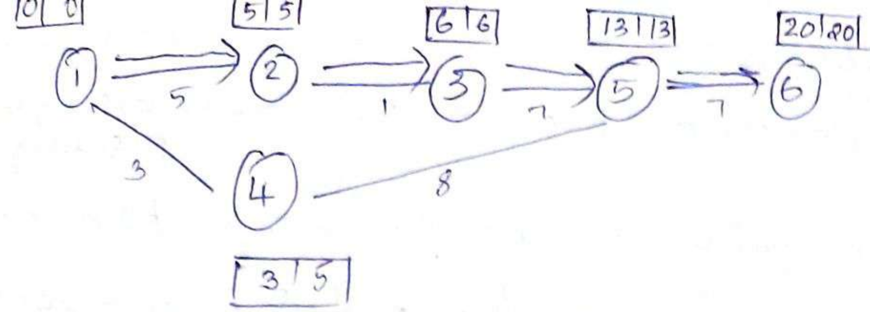
ⓒ Calculate EST & LFT and mark on network diagram

ⓓ Calculate the total slack for each activity

ⓔ Identify the Critical path and mark on network diagram

**Ans**

| Activity | to | tm | tp | te | $\sigma$ | $\sigma^2$ | EST | LFT | LST | LFT | F |
|---|---|---|---|---|---|---|---|---|---|---|---|
| 1 – 2 | 2 | 5 | 8 | 5 | 1 | 1 | 0 | 5 | 0 | 5 | 0 |
| 2 – 3 | 1 | 1 | 1 | 1 | 0 | 0 | 5 | 6 | 5 | 6 | 0 |
| 3 – 5 | 0 | 6 | 18 | 7 | 3 | 9 | 6 | 13 | 6 | 13 | 0 |
| 5 – 6 | 7 | 7 | 7 | 7 | 0 | 0 | 20 | 20 | 13 | 20 | 0 |
| 1 – 4 | 3 | 3 | 3 | 3 | 0 | 0 | [illegible] | 3 | 2 | 5 | 2. |
| 4 – 5 | 2 | 8 | 14 | 8 | 2 | 4 | 3 | 11 | 5 | 13 | 2 |



EST → write Maximum

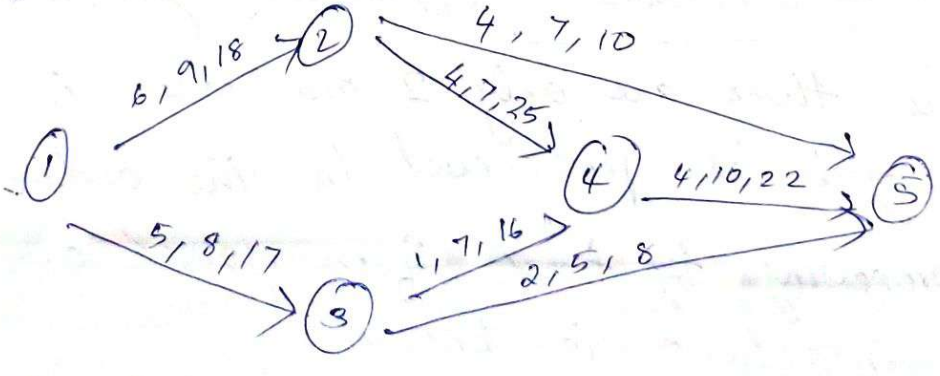
① —1— ②

| EST | LFT |

| | | | |
|---|---|---|---|
| 1 - 2 | 0 \| 5 | 0 \| 5 | 0 |
| 2 - 3 | 5 \| 6 | 5 \| 6 | 0 |
| 3 - 5 | 6 \| 13 | 6 \| 13 | 0 |
| 5 - 6 | 13 \| 20 | 13 \| 20 | 0 |
| 1 - 4 | 0 \| 3 | 2 \| 5 | 2 |
| 4 - 5 | 3 \| 11 | 5 \| 13 | 2 |

HW ② For the network shown below the time estimate (in days) each for activity are indicated on the diagram ① determine the expected time & variance for each activity
② the probability of completing the project in 32 days
③ Total project duration
④ Mark Critical path.

[Alway they give in this order to tm tp]



$\cancel{z = \frac{t_f - te}{6}}$

$$Z = \frac{Ts - Te}{\sigma}$$

# MODULE – 3

## DIFFERENT CONTROL METHOD :-

loading : Assigning a Specific job/work to a machine or work center for the planning period.

Sequencing : Determining the order of processing of all jobs at each work center.
(Problem using Johnsons Algorithm)

Scheduling : The process of establishing start and finish time of all jobs at each work center.

## Sequencing Problems

The sequencing problems arise whenever there is a need for determining an optimum order of performing a no: of jobs by no of faculty according to some pre assigned order so as to optimise the output in terms of cost, time or profit

### types

(1) 'n' jobs and 2 Machines
(2) 'n' jobs and 3 Machines
(3) 'n' jobs and 'm' Machines

### (1) → n jobs & 2 Machine (Johnsons Algorithm),

In this there are only 2 machines A & B, each job is processed in the order A & B. The processing ~~tg~~ times of n jobs on each of the 2 machine is known

(1) Select the smallest processing time from the given list of processing time.
(2) If the minimum processing time is A (Job no: r on the machine A) do the rth job first in the

sequence. if the minimum processing time is B do that job last in the sequence.

③ After doing this step, n-1 jobs are left to sequenced, repeat step 1 and 2 till all the jobs are orderd

④ find the total processing time as per the sequence and also determine Ideal time associated with machines

Q, five jobs are to be proceesed on 2 machines M1 & M2 In the order M,M. Processing time in hrs are given betermine the sequence that minimise total elapsed time. find out the total elapsed time and ideal time on M2

| Job | M/C 1 | M/C 2 |
|---|---|---|
| | Time hrs | |
| 1 | 5 | 2 |
| 2 | 1 | 6 |
| 3 | 9 | 7 |
| 4 | 3 | 8 |
| 5 | 10 | 4 |

| | | | | |
|---|---|---|---|---|
| 2 | 4 | 3 | 5 | 1 |

here 1 is minimum and since in M1 write in first colum.
next 2 is minimum and since in M2 write job1 last similarly do.

total elapsed time.

~~Optimal sequence M/C 1 M/C 2 In out Idle time~~

| optimal sequence | MC1 In | MC1 out | MC2 In | MC2 out | Idle time |
|---|---|---|---|---|---|
| 2 | 0 | 1 | 1 | 7 | 1 |
| 4 | 1 | 4 | 7 | 15 | 0 |
| 3 | 4 | 13 | 15 | 22 | 0 |
| 5 | 13 | 23 | 23 | 27 | 1 |
| 1 | 23 | 28 | 28 | 30 | 1 |

3hrs

(2) There are 7 jobs which are to be processed first on machine 1 and then on 2, processing time in hrs are given below find optimum sequence and total elapsed time and Idle time of $M_2$

| Job | M/c 1 | M/c 2 |
|---|---|---|
| A | 6 | 16 |
| B | 24 | 20 |
| C | 30 | 20 |
| D | 12 | 13 |
| E | 20 | 24 |
| F | 22 | 2 |
| G | 18 | 6 |

| A | D | E | C | B | G | F |
|---|---|---|---|---|---|---|

| Optimum sequence | Mc In | Mc out | MC 2 In | MC 2 out | Idle time |
|---|---|---|---|---|---|
| A | 0 | 6 | | | |
| D | 6 | 18 | | | |
| E | 18 | 38 | | | |
| C | 38 | 68 | | | |
| B | 68 | 92 | | | |
| G | 92 | 110 | | | |
| F | 110 | 132 | | | |

Minimum elapsed time ~~103~~ 134 hrs
Idle time on 2 = 33 hrs//